

Taller 1

```
Usuario@DESKTOP-8MC9K4I MINGW64 ~
$ ls -la saludo.sh monitor.sh analiza.sh access.log reporte.txt
-rw-r--r-- 1 Usuario 197121 455 Aug 30 17:59 access.log
-rwxr-xr-x 1 Usuario 197121 874 Aug 30 18:19 analiza.sh*
-rwxr-xr-x 1 Usuario 197121 570 Aug 30 18:25 monitor.sh*
-rw-r--r-- 1 Usuario 197121 187 Aug 30 18:01 reporte.txt
-rwxr-xr-x 1 Usuario 197121 111 Aug 30 18:10 saludo.sh*

Usuario@DESKTOP-8MC9K4I MINGW64 ~
$ ./saludo.sh
echo $?
Hola desde mi primer script Bash
Usuario: Usuario
Directorio actual: /c/Users/Usuario
0

Usuario@DESKTOP-8MC9K4I MINGW64 ~
$ ./monitor.sh
===== MONITOR DEL CONTENEDOR =====
Hostname: DESKTOP-8MC9K4I
Usuario: Usuario
Fecha: Sun Aug 30 18:25:44 HPS 2026
Directorio: /c/Users/Usuario
===== DISCO =====
Filesystem      Size  Used Avail Use% Mounted on
C:/Program Files/Git  476G  398G   79G  84% /
D:                932G   25G  907G   3% /d
./monitor.sh: line 14: [: 79G: integer expression expected
===== PROCESOS =====
Procesos activos aproximados: 7
===== FIN DEL REPORTE =====
El diagnostico termino correctamente.

Usuario@DESKTOP-8MC9K4I MINGW64 ~
$ ./analiza.sh
./analiza.sh no_existe.log
./analiza.sh access.log
Error: Debe proporcionar al menos un archivo como argumento.
Uso: ./analiza.sh <archivo1> [archivo2 ...]
-----
Error: El archivo 'no_existe.log' no existe.
-----
Analizando archivo: access.log
Cantidad de lineas: 7
Cantidad de palabras: 42
Cantidad de caracteres: 455
Lineas con ERROR: 3
Lineas con WARN: 1
-----

Usuario@DESKTOP-8MC9K4I MINGW64 ~
$ cat reporte.txt
===== REPORTE DE LOGS =====
1. Total de registros: 7
2. Cantidad de errores: 3
3. Cantidad de advertencias: 1
4. Usuario con mas errores: user=juan
5. IP con mas errores: ip=192.168.1.15
```

Taller 2 Parte 1

```

Usuario@DESKTOP-8MC9K4I MINGW64 ~/taller2_docker
$ cat > Dockerfile << 'EOF'
FROM ubuntu:latest
CMD [ "echo", "Hola desde CMD" ]
EOF

docker build -t ejemplo-cmd .
docker run --rm ejemplo-cmd
docker run --rm ejemplo-cmd echo "Mensaje reemplazado"
[+] Building 0.7s (5/5) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 89B                               0.0s
=> [internal] load metadata for docker.io/library/ubuntu:latest 0.4s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                     0.0s
=> CACHED [1/1] FROM docker.io/library/ubuntu:latest@sha256:2260313b31c8 0.0s
=> => resolve docker.io/library/ubuntu:latest@sha256:2260313b31c8c011cd2 0.0s
=> exporting to image                                           0.1s
=> => exporting layers                                           0.0s
=> => exporting manifest sha256:3ed3aa4d84c511c9c3bf3da13d378c368476a327 0.0s
=> => exporting config sha256:bf8cb9f21a90fd894a7f4c0b2ca20fd31f39677463 0.0s
=> => exporting attestation manifest sha256:47ef6ee025b582dc36aad9f8f0d7 0.0s
=> => exporting manifest list sha256:35e54e6475300bf7c87bec56488d1ff4f74 0.0s
=> => naming to docker.io/library/ejemplo-cmd:latest            0.0s
=> => unpacking to docker.io/library/ejemplo-cmd:latest         0.0s
Hola desde CMD
Mensaje reemplazado

Usuario@DESKTOP-8MC9K4I MINGW64 ~/taller2_docker
$ cat > Dockerfile << 'EOF'
FROM ubuntu:latest
ENTRYPOINT [ "echo" ]
EOF

docker build -t ejemplo-entrypoint .
docker run --rm ejemplo-entrypoint "Hola"
docker run --rm ejemplo-entrypoint "Mensaje adicional"
[+] Building 1.1s (5/5) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 78B                               0.0s
=> [internal] load metadata for docker.io/library/ubuntu:latest 0.3s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                     0.0s
=> CACHED [1/1] FROM docker.io/library/ubuntu:latest@sha256:2260313b31c8 0.1s
=> => resolve docker.io/library/ubuntu:latest@sha256:2260313b31c8c011cd2 0.1s
=> exporting to image                                           0.3s
=> => exporting layers                                           0.0s
=> => exporting manifest sha256:e98a316fa79fd45fd55f4a89c65093ae3375bbf5 0.1s
=> => exporting config sha256:73228b9a057d67d302bb2bf72144651db07dc5d1ba 0.1s
=> => exporting attestation manifest sha256:26d9ad6ff8a472e019fbec47651f 0.1s
=> => exporting manifest list sha256:3f78df5f78bd631844bf50b57189e85e0c 0.1s
=> => naming to docker.io/library/ejemplo-entrypoint:latest    0.0s
=> => unpacking to docker.io/library/ejemplo-entrypoint:latest 0.0s
Hola
Mensaje adicional

Usuario@DESKTOP-8MC9K4I MINGW64 ~/taller2_docker
$ cat > Dockerfile << 'EOF'
FROM ubuntu:latest
ENTRYPOINT [ "echo" ]
CMD [ "Mensaje por defecto" ]
EOF

docker build -t ejemplo-combinado .
docker run --rm ejemplo-combinado
docker run --rm ejemplo-combinado "Mensaje personalizado"
[+] Building 1.1s (5/5) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 108B                               0.0s
=> [internal] load metadata for docker.io/library/ubuntu:latest 0.2s

```

Taller 2 parte 2

```

Usuario@DESKTOP-8MC9K4I MINGW64 ~/taller2_docker
$ cat > Dockerfile << 'EOF'
FROM ubuntu:latest
ENTRYPOINT [ "echo" ]
CMD [ "Mensaje por defecto" ]
EOF

docker build -t ejemplo-combinado .
docker run --rm ejemplo-combinado
docker run --rm ejemplo-combinado "Mensaje personalizado"
[+] Building 1.1s (5/5) FINISHED          docker:desktop-linux
=> [internal] load build definition from Dockerfile          0.0s
=> => transferring dockerfile: 108B                          0.0s
=> [internal] load metadata for docker.io/library/ubuntu:latest 0.2s
=> [internal] load .dockerignore                             0.0s
=> => transferring context: 2B                                0.0s
=> CACHED [1/1] FROM docker.io/library/ubuntu:latest@sha256:2260313b31c8 0.1s
=> => resolve docker.io/library/ubuntu:latest@sha256:2260313b31c8c011cd2 0.1s
=> exporting to image                                         0.3s
=> => exporting layers                                         0.0s
=> => exporting manifest sha256:d4f9e5e34f52fae5eefa05060fd9153d58476371 0.1s
=> => exporting config sha256:b139486132307837949646b89890488e6ef915bedf 0.1s
=> => exporting attestation manifest sha256:083ea13efa0d77fdc579e4b94ac2 0.1s
=> => exporting manifest list sha256:36913b5f7c46ba9e26cbfef9813ea17787f 0.1s
=> => naming to docker.io/library/ejemplo-combinado:latest 0.0s
=> => unpacking to docker.io/library/ejemplo-combinado:latest 0.0s
Mensaje por defecto
Mensaje personalizado

```

Archivo Docker file

FROM ubuntu:latest

ENTRYPOINT ["echo"]

CMD ["Mensaje por defecto"]

Conclusión

El uso combinado de CMD y ENTRYPOINT en la construcción de imágenes de Docker permite definir de forma clara el propósito y la flexibilidad de un contenedor. Mientras que ENTRYPOINT establece el ejecutable o proceso principal que no debe modificarse durante la ejecución, CMD proporciona los parámetros o comandos por defecto que el usuario puede sobrescribir fácilmente desde la terminal al ejecutar Docker run. Así, mientras que CMD ofrece flexibilidad total y ENTRYPOINT garantiza una herramienta fija, la unión de ambas instrucciones constituye la mejor práctica para crear aplicaciones parametrizables que mantengan un binario base definido, pero permitan ajustar su comportamiento mediante argumentos dinámicos.

Taller 3 parte 1-2

```

Usuario@DESKTOP-8MC9K4I MINGW64 ~
$ mkdir -p ~/taller3_compose && cd ~/taller3_compose

Usuario@DESKTOP-8MC9K4I MINGW64 ~/taller3_compose
$ cat > docker-compose.yml << 'EOF'
services:
  web:
    image: nginx:alpine
    ports:
      - "8080:80"
EOF

Usuario@DESKTOP-8MC9K4I MINGW64 ~/taller3_compose
$ docker compose up -d
[+] up 13/13
  ✓ Image nginx:alpine Pulled 4.8s
  ✓ Network taller3_compose_default Created 0.0s
  ✓ Container taller3_compose-web-1 Started 0.5s

Usuario@DESKTOP-8MC9K4I MINGW64 ~/taller3_compose
$ docker compose ps
docker compose logs web

```

NAME	STATUS	IMAGE	COMMAND	SERVICE	CREATE
taller3_compose-web-1	Up 11 seconds	nginx:alpine	"/docker-entrypoint..."	web	11 sec

```

web-1 | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt
web-1 | to perform configuration
web-1 | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.
web-1 | d/
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6
web-1 | -by-default.sh
web-1 | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/ngi
web-1 | nx/conf.d/default.conf
web-1 | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/n
web-1 | ginx/conf.d/default.conf
web-1 | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers
web-1 | .envsh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templ
web-1 | ates.sh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-pro
web-1 | cesses.sh
web-1 | /docker-entrypoint.sh: Configuration complete; ready for start up
web-1 | 2026/08/31 00:17:56 [notice] 1#1: using the "epoll" event method
web-1 | 2026/08/31 00:17:56 [notice] 1#1: nginx/1.31.4
web-1 | 2026/08/31 00:17:56 [notice] 1#1: built by gcc 15.2.0 (Alpine 15.2.0)
web-1 | 2026/08/31 00:17:56 [notice] 1#1: OS: Linux 6.18.33.2-microsoft-standar
web-1 | d-WSL2
web-1 | 2026/08/31 00:17:56 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:104
web-1 | 8576
web-1 | 2026/08/31 00:17:56 [notice] 1#1: start worker processes
web-1 | 2026/08/31 00:17:56 [notice] 1#1: start worker process 30
web-1 | 2026/08/31 00:17:56 [notice] 1#1: start worker process 31
web-1 | 2026/08/31 00:17:56 [notice] 1#1: start worker process 32
web-1 | 2026/08/31 00:17:56 [notice] 1#1: start worker process 33
web-1 | 2026/08/31 00:17:56 [notice] 1#1: start worker process 34
web-1 | 2026/08/31 00:17:56 [notice] 1#1: start worker process 35
web-1 | 2026/08/31 00:17:56 [notice] 1#1: start worker process 36
web-1 | 2026/08/31 00:17:56 [notice] 1#1: start worker process 37
web-1 | 2026/08/31 00:17:56 [notice] 1#1: start worker process 38
web-1 | 2026/08/31 00:17:56 [notice] 1#1: start worker process 39
web-1 | 2026/08/31 00:17:56 [notice] 1#1: start worker process 40
web-1 | 2026/08/31 00:17:56 [notice] 1#1: start worker process 41

Usuario@DESKTOP-8MC9K4I MINGW64 ~/taller3_compose
$ docker compose down
[+] down 2/2
  ✓ Container taller3_compose-web-1 Removed 3.4s
  ✓ Network taller3_compose_default Removed 0.2s

```

```
Usuario@DESKTOP-8MC9K4I MINGW64 ~/taller3_compose
$ mkdir -p html
cat > html/index.html << 'EOF'
<!DOCTYPE html>
<html lang="es">
<head>
  <meta charset="UTF-8">
  <title>Mi primer servicio Docker</title>
</head>
<body>
  <h1>Hola desde Nginx y Docker Compose</h1>
  <p>Este contenido está siendo servido desde un contenedor.</p>
</body>
</html>
EOF

Usuario@DESKTOP-8MC9K4I MINGW64 ~/taller3_compose
$ cat > docker-compose.yml << 'EOF'
services:
  web:
    image: nginx:alpine
    ports:
      - "8080:80"
    volumes:
      - ./html:/usr/share/nginx/html:ro
EOF

Usuario@DESKTOP-8MC9K4I MINGW64 ~/taller3_compose
$ docker compose up -d
[+] up 2/2
✓Network taller3_compose_default Created
✓Container taller3_compose-web-1 Started
```

Hola desde Nginx y Docker Compose

Este contenido está siendo servido desde un contenedor.

¡Edición en tiempo real comprobada!

Taller 3 parte 3

```
Usuario@DESKTOP-8MC9K4I MINGW64 ~/taller3_compose
$ cat > html/index.html << 'EOF'
<!DOCTYPE html>
<html lang="es">
<head>
  <meta charset="UTF-8">
  <title>Mi primer servicio Docker</title>
</head>
<body>
  <h1>Hola desde Nginx y Docker Compose</h1>
  <p>Este contenido está siendo servido desde un contenedor.</p>
  <p>¡Edición en tiempo real comprobada!</p>
</body>
</html>
EOF
```

```
Usuario@DESKTOP-8MC9K4I MINGW64 ~/taller3_compose
$ cat > Dockerfile << 'EOF'
FROM nginx:alpine
COPY html/ /usr/share/nginx/html/
EOF
```

```
Usuario@DESKTOP-8MC9K4I MINGW64 ~/taller3_compose
$ cat > docker-compose.yml << 'EOF'
services:
  web:
    build: .
    ports:
      - "8080:80"
EOF
```

```

Usuario@DESKTOP-8MC9K41 MINGW64 ~/taller3_compose
$ docker compose down
docker compose up --build -d
[+] down 2/2
  ✓ Container taller3_compose-web-1 Removed 0.5s
  ✓ Network taller3_compose_default Removed 0.2s
#1 [internal] load local bake definitions
#1 reading from stdin 527B 0.0s done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 89B done
#2 DONE 0.1s

#3 [internal] load metadata for docker.io/library/nginx:alpine
#3 DONE 0.1s

#4 [internal] load .dockerignore
#4 transferring context:
#4 transferring context: 2B done
#4 DONE 0.1s

#5 [1/2] FROM docker.io/library/nginx:alpine@sha256:db35bfc6b2951e7f8a72db5db120
288c127ffaeb4a6d4b95a26fead017d5913
#5 resolve docker.io/library/nginx:alpine@sha256:db35bfc6b2951e7f8a72db5db120288
c127ffaeb4a6d4b95a26fead017d5913 0.1s done
#5 ...

#6 [internal] load build context
#6 transferring context: 379B done
#6 DONE 0.2s

#5 [1/2] FROM docker.io/library/nginx:alpine@sha256:db35bfc6b2951e7f8a72db5db120
288c127ffaeb4a6d4b95a26fead017d5913
#5 DONE 0.7s

#7 [2/2] COPY html/ /usr/share/nginx/html/
#7 DONE 0.0s

#8 exporting to image
#8 exporting layers 0.1s done
#8 exporting manifest sha256:93e412f86a82da4f2b3bce9c74e66543b4338053f9fc4b28856
6ad7f26b0fc8b 0.0s done
#8 exporting config sha256:a6e4dffe69d5cbb9d7561c2b1669ab7bc4bdc7d29bf1d2c317f47
d9654bd0607 0.0s done
#8 exporting attestation manifest sha256:2d62a9ab8c7600b318c0917430b36898b4dc3c8
d6a3ca0032910c987203f1dcc 0.0s done
#8 exporting manifest list sha256:729e177d330b778fd44e57734d7abbe2fbe158d11224fe
dc6164b6652b5c3743 0.0s done
#8 naming to docker.io/library/taller3_compose-web:latest done
#8 unpacking to docker.io/library/taller3_compose-web:latest 0.0s done
#8 DONE 0.2s

#9 resolving provenance for metadata file
#9 DONE 0.0s
[+] up 3/3
  ✓ Image taller3_compose-web Built 1.8s
  ✓ Network taller3_compose_default Created 0.0s
  ✓ Container taller3_compose-web-1 Started 0.3s

```

Taller 3 parte 4

```
Usuario@DESKTOP-8MC9K4I MINGW64 ~/taller3_compose
$ mkdir -p scripts logs

cat > scripts/monitor.sh << 'EOF'
#!/bin/bash
echo "=== MONITOR DEL CONTENEDOR ==="
echo "Fecha: $(date)" >> /app/logs/reporte.txt
echo "Hostname: $(hostname)" >> /app/logs/reporte.txt
echo "Monitoreo completado."
EOF

cat > scripts/analiza.sh << 'EOF'
#!/bin/bash
echo "=== ANALIZANDO LOGS ==="
if [ -f "/app/logs/reporte.txt" ]; then
    cat /app/logs/reporte.txt
else
    echo "No se encontro archivo de reporte."
fi
EOF

chmod +x scripts/*.sh

Usuario@DESKTOP-8MC9K4I MINGW64 ~/taller3_compose
$ cat > docker-compose.yml << 'EOF'
services:
  web:
    build: .
    ports:
      - "8080:80"
    volumes:
      - shared-logs:/usr/share/nginx/html/logs

  ubuntu:
    image: ubuntu:latest
    command: tail -f /dev/null
    volumes:
      - ./scripts:/app/scripts
      - shared-logs:/app/logs

volumes:
  shared-logs:
EOF
```

```

Usuario@DESKTOP-8MC9K4I MINGW64 ~/taller3_compose
$ docker compose up --build -d
[+] up 1/1
  Image ubuntu:latest Pulled 2.8s
#1 [internal] load local bake definitions
#1 reading from stdin 527B 0.0s done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 89B done
#2 DONE 0.0s

#3 [internal] load metadata for docker.io/library/nginx:alpine
#3 DONE 0.0s

#4 [internal] load .dockerignore
#4 transferring context: 2B done
#4 DONE 0.0s

#5 [internal] load build context
#5 transferring context: 61B done
#5 DONE 0.0s

#6 [1/2] FROM docker.io/library/nginx:alpine@sha256:db35bfc6b2951e7f8a72db5db120
288c127ffaeb4a6d4b95a26fead017d5913
#6 resolve docker.io/library/nginx:alpine@sha256:db35bfc6b2951e7f8a72db5db120288
c127ffaeb4a6d4b95a26fead017d5913 0.0s done
#6 DONE 0.0s

#7 [2/2] COPY html/ /usr/share/nginx/html/
#7 CACHED

#8 exporting to image
#8 exporting layers done
#8 exporting manifest sha256:93e412f86a82da4f2b3bce9c74e66543b4338053f9fc4b28856
6ad7f26b0fc8b done
#8 exporting config sha256:a6e4dffe69d5cbb9d7561c2b1669ab7bc4bdc7d29bf1d2c317f47
d9654bd0607 done
#8 exporting attestation manifest sha256:3329d9ee2c3220be1b98fe62895dc38793ddab3
3fd709478a276e098dadc4f7d 0.0s done
#8 exporting manifest list sha256:64e6888a7538f53caeb086671c20612a773a24e63078ac
f58b2198221e4080ea
#8 exporting manifest list sha256:64e6888a7538f53caeb086671c20612a773a24e63078ac
f58b2198221e4080ea 0.0s done
#8 naming to docker.io/library/taller3_compose-web:latest done
#8 unpacking to docker.io/library/taller3_compose-web:latest 0.0s done
#8 DONE 0.1s

[+] up 5/5ng provenance for metadata file
  Image ubuntu:latest Pulled 2.8s
  Image taller3_compose-web Built 0.6s
  Volume taller3_compose_shared-logs Created 0.0s
  Container taller3_compose-web-1 Started 3.7s
  Container taller3_compose-ubuntu-1 Started 3.8s

```

```

Usuario@DESKTOP-8MC9K4I MINGW64 ~/taller3_compose
$ docker compose exec ubuntu /app/scripts/monitor.sh
docker compose exec ubuntu /app/scripts/analiza.sh
OCI runtime exec failed: exec failed: unable to start container process: exec: "
C:/Program Files/Git/app/scripts/monitor.sh": stat C:/Program Files/Git/app/scri
pts/monitor.sh: no such file or directory

```

What's next:

Debug this Compose error with Gordon → docker ai "help me fix this compose e rror"

```

OCI runtime exec failed: exec failed: unable to start container process: exec: "
C:/Program Files/Git/app/scripts/analiza.sh": stat C:/Program Files/Git/app/scri
pts/analiza.sh: no such file or directory

```

What's next:

Debug this Compose error with Gordon → docker ai "help me fix this compose e rror"

Archivos de código fuente

Docker compose.yml

```
services:
  web:
    build: .
    ports:
      - "8080:80"
    volumes:
      - shared-logs:/usr/share/nginx/html/logs
  ubuntu:
    image: ubuntu:latest
    command: tail -f /dev/null
    volumes:
      - ./scripts:/app/scripts
      - shared-logs:/app/logs
volumes:
  shared-logs:
```

Dockerfile

```
FROM nginx:alpine
COPY html/ /usr/share/nginx/html/
```

Monitor.sh

```
#!/bin/bash
echo "=== MONITOR DEL CONTENEDOR ==="
echo "Fecha: $(date)" >> /app/logs/reporte.txt
echo "Hostname: $(hostname)" >> /app/logs/reporte.txt
echo "Monitoreo completado."
```

Scripts/analiza.sh

```
#!/bin/bash
```

```
echo "=== ANALIZANDO LOGS ==="
if [ -f "/app/logs/reporte.txt" ]; then
    cat /app/logs/reporte.txt
else
    echo "No se encontro archivo de reporte."
fi
```

Preguntas Parte 2

¿Por qué el cambio en index.html aparece sin reconstruir la imagen?

Porque la carpeta. /html está montada mediante un volumen de tipo bind mount. Nginx lee directamente los archivos desde el sistema de archivos de la máquina host en tiempo real.

¿Qué función cumple: ro?

Indica que el volumen se monta en modo solo lectura (Read-Only). Esto impide que el contenedor modifique, cree o elimine archivos dentro de la carpeta host original.

Diferencia entre el puerto 8080 y el 80 ("8080:80"):

El puerto 8080 corresponde a la máquina host (tu computadora), mientras que el puerto 80 es el puerto interno dentro del contenedor donde Nginx escucha las peticiones.

Identificación de Host vs. Contenedor:

En los puertos "8080:80", 8080 es del host y 80 del contenedor.

En los volúmenes. /html:/usr/share/nginx/html:ro, /html es la ruta local del host y /usr/share/nginx/html es la ruta dentro del contenedor.

Pregunta de Diseño

Docker Compose permite definir la arquitectura completa de una aplicación (servicios, redes, volúmenes compartidos y puertos) de forma declarativa dentro de un único archivo docker-compose.yml. Esto evita la necesidad de recordar y ejecutar comandos Docker run extensos y propensos a errores, permitiendo gestionar el ciclo de vida completo de la infraestructura con comandos simples como Docker compose up y Docker compose Down.